

eSecurity – AS 9626

1. Nov/2021/Paper_11/No.3

Many online banking applications use the HTTPS protocol, involving encryption, for secure connections.

Evaluate, by weighing up the advantages and disadvantages, the use of HTTPS and the need for encryption in banking websites.

[8]

2. Nov/2021/Paper_11/No.7

A company has several IT workers. To help reduce health issues the company encourages workers to take regular breaks.

Describe, in detail, **three** different health issues that can affect these workers. For each issue give one method, other than taking regular breaks, of helping to prevent it. Each method must be different.

Issue 1

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Prevention

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Issue 2

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Issue 3

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Prevention

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3. Nov/2021/Paper_11/No.8

Many companies have branches in different cities. Branch workers can connect to the main office using the company's virtual private network (VPN) connection.

Describe the characteristics of this type of VPN.

[6]

4. Nov/2021/Paper_12/No.6

Personal data should always be kept confidential and secure. Data can be kept secure by using encryption and by restricting access using methods such as passwords and biometrics.

Describe other methods of keeping data confidential and secure.

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5. Nov/2021/Paper_13/No.3

Fraudsters sometimes attempt to obtain customers' personal and banking data so they can commit identity theft.

Describe how customers should try to prevent their personal and banking data being obtained by phishing.

[6]

6. Nov/2021/Paper_13/No.4

Many parents are concerned that their children are not aware of the e-safety issues that arise from using the internet.

Describe the guidance parents should give to make sure their child stays e-safe when using social networking and gaming sites.

[6]

7. June/2021/Paper_12/No.4

Define the following types of malware, including a description of what each type does.

(a) Worm

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(b) Spyware

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8. June/2021/Paper_13/No.1c

- (c) The SSL/TLS protocol is an encryption protocol frequently used when data is transmitted over the internet.

Describe how the SSL/TLS protocol is used to ensure the security of the data.

[6]

9. June/2021/Paper_13/No.5

A company has several IT rooms. The computers in each IT room are connected to the company's network. One of the physical safety issues is workers tripping over cables.

Describe, in detail, **three** other physical safety issues that can arise from the use of IT rooms. For each issue, give one method of helping to prevent it. Each method must be different.

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Prevention method

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Issue 2

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Prevention method

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Issue 3

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Prevention method

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10. Nov/2020/Paper_11/No.7

Define the following types of malware including a description of what each type does.

(a) Rootkit

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(b) Malicious bots

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(c) Ransomware

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11. Nov/2020/Paper_12/No.1

(a) Tick the most accurate statement regarding the use of vishing.

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Vishing involves mobile (cell) phone text messages persuading people to reveal their personal data	
With vishing, a text message may include a telephone number that connects to an automated voice response system	
Vishing involves installing a piece of malicious software on a customer's computer	
Vishing involves a phone call to persuade people to reveal their personal data	

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(b) Describe, in detail, **three** measures you could take to help prevent you being a victim of vishing.

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(c) Tick the most accurate statement regarding the use of smishing.

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With smishing, a text message is received which may include a website URL and invite the receiver to go to the site	
Smishing is the use of video clips persuading people to reveal their personal data	
Smishing involves using emails to persuade people to reveal their personal data	
Smishing involves receiving a phone call	

[1]

12. Nov/2020/Paper_13/No.7

Define the following types of malware including a description of what each type does.

(a) Rootkit

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(b) Malicious bots

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(c) Ransomware

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13. March/2020/Paper_12/No.8

Describe the features of the encryption protocols TLS and SSL.

[6]

14. Mar/2020/Paper_12/No.10

In a recent BBC news bulletin it was said:

'It takes 82 seconds for cyber-thieves to ensnare the first victim of a phishing campaign, a report suggests.'

'Hackers exploit router flaws in unusual pharming attack.'

Describe what is meant by:

(a) phishing

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(b) pharming.

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15. June/2020/Paper_11/No.8

Explain how encryption protects data stored on a hard disk.

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Describe the advantages of a school having an e-safety policy.

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17. June/2020/Paper_12/No.6

Personal data and other sensitive types of data often need to be kept confidential or restricted to authorised staff.

Explain why personal data should be kept confidential.

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18. June/2020/Paper_12/No.8

Describe how **two** different safety issues can arise from the use of IT. For each issue give **one** method of helping to prevent it. Each method must be different.

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19. June/2020/Paper_13/No.9

There are two types of encryption, symmetric and asymmetric.

- (a) Describe the use of public keys and private keys when used in the asymmetric method of encryption.

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- (b) Apart from the types of keys used, give **two** reasons why symmetric encryption is preferred in certain situations.

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